Light

GCSE Computer Science — CPU Architecture (Set 1)

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

A computer is performing a calculation to work out the total cost of items in a shopping basket.
Which component of the CPU performs this calculation?

- ○ **A.** The Arithmetic Logic Unit (ALU)
- ○ **B.** The Control Unit (CU)
- ○ **C.** The Program Counter (PC)
- ○ **D.** The Memory Data Register (MDR)

Q2 (1 mark)

A program is being executed on a computer. The CPU needs to know which instruction to fetch next from memory.

Which register holds the memory address of the next instruction to be fetched?

- ○ **A.** The Memory Address Register (MAR)
- ○ **B.** The Memory Data Register (MDR)
- ○ **C.** The Program Counter (PC)
- ○ **D.** The Accumulator (ACC)

Q3 (1 mark)

During the fetch-decode-execute cycle, one component of the CPU is responsible for coordinating all operations and sending control signals to other components.
Which component has this role?

- ○ **A.** The Arithmetic Logic Unit (ALU)
- ○ **B.** The Accumulator (ACC)
- ○ **C.** The Clock
- ○ **D.** The Control Unit (CU)

Q4 (1 mark)

A computer has three main buses connecting the CPU to memory and other components.
Which bus carries the memory addresses that identify where data should be read from or written to?

- ○ **A.** The data bus
- ○ **B.** The address bus

- ○ **C.** The control bus
- ○ **D.** The system bus

Q5 (1 mark)

During the fetch-decode-execute cycle, the CPU needs to tell RAM whether it wants to read data or write data.
Which bus carries this read/write signal?

- ○ **A.** The data bus
- ○ **B.** The address bus
- ○ **C.** The control bus
- ○ **D.** The system bus

Q6 (1 mark)

When data is being transferred between the CPU and RAM, it must be temporarily held in a register.
Which register temporarily holds data that is being transferred to or from memory?

- ○ **A.** The Program Counter (PC)
- ○ **B.** The Current Instruction Register (CIR)
- ○ **C.** The Memory Address Register (MAR)
- ○ **D.** The Memory Data Register (MDR)

Q7 (1 mark)

A CPU contains a component that generates regular electrical pulses to synchronise all operations.
What is this component called?

- ○ **A.** The clock
- ○ **B.** The oscillator
- ○ **C.** The timer
- ○ **D.** The pulse generator

Q8 (1 mark)

The three main buses in a computer system can be either unidirectional (signals travel in one direction only) or bidirectional (signals can travel in both directions).
Which bus is unidirectional, carrying signals only from the CPU to memory?

- ○ **A.** The data bus
- ○ **B.** The control bus
- ○ **C.** The address bus
- ○ **D.** The system bus

Q9 (1 mark)

During the fetch-decode-execute cycle, the instruction currently being processed must be stored somewhere while it is decoded.
Which register holds the instruction that is currently being decoded and executed?

- ○ **A.** The Current Instruction Register (CIR)

- ○ **B.** The Program Counter (PC)
- ○ **C.** The Accumulator (ACC)
- ○ **D.** The Memory Address Register (MAR)

Q10 (1 mark)

The CPU processes instructions using a continuous cycle of three stages.
What is the first stage of this cycle?

- ○ **A.** Fetch
- ○ **B.** Execute
- ○ **C.** Decode
- ○ **D.** Store

Q11 (1 mark)

After an instruction has been fetched from memory, it must be interpreted to determine what action the CPU should take.
Which component decodes the instruction?

- ○ **A.** The Arithmetic Logic Unit (ALU)
- ○ **B.** The Memory Data Register (MDR)
- ○ **C.** The Control Unit (CU)
- ○ **D.** The Program Counter (PC)

Q12 (1 mark)

During a series of calculations, the CPU needs to store intermediate results before the final answer is ready.
Which register typically stores these intermediate calculation results?

- ○ **A.** The Memory Address Register (MAR)
- ○ **B.** The Program Counter (PC)
- ○ **C.** The Current Instruction Register (CIR)
- ○ **D.** The Accumulator (ACC)

Q13 (2 marks)

A computer manufacturer releases two versions of a laptop. Both have identical specifications except for the CPU clock speed:
Model A: 2.4 GHz clock speed
Model B: 3.2 GHz clock speed
Which statement correctly explains how clock speed affects CPU performance?

- ○ **A.** Model B can execute more instructions per second because each clock cycle triggers a new step in the fetch-decode-execute cycle.
- ○ **B.** Model B uses more electricity because higher clock speeds always require additional power circuits.
- ○ **C.** Model A is faster because lower clock speeds generate less heat, allowing the CPU to work more efficiently.
- ○ **D.** Model B stores more data in its registers because higher clock speeds require larger memory capacity.

Q14 (2 marks)

During the fetch stage of the FDE cycle, the following operations occur:
1. The address of the next instruction is sent to RAM
2. The instruction is transferred from RAM to the CPU
3. The Program Counter is incremented
Which registers are involved in operation 1?

- ○ **A.** The PC copies its value to the MAR, then the MAR sends the address via the address bus
- ○ **B.** The MDR sends the address directly to RAM via the data bus
- ○ **C.** The CIR stores the address and sends it to RAM via the control bus
- ○ **D.** The ACC calculates the address and sends it to the MAR

Q15 (2 marks)

A computer system has a 16-bit address bus.
Which statement correctly explains a consequence of this address bus width?

- ○ **A.** The CPU can address a maximum of 65,536 different memory locations.
- ○ **B.** The system can have a maximum of 16 memory locations.
- ○ **C.** The CPU can transfer 16 bits of data to memory in each clock cycle.
- ○ **D.** The system can only run programs that are 16 bits in size.

Q16 (2 marks)

During the fetch stage of the fetch-decode-execute cycle, several operations occur involving different registers.
Which sequence correctly describes what happens during the fetch stage?

○ **A.** The instruction in the CIR is decoded, then the ALU performs the required calculation, then results are stored in the Accumulator.

○ **B.** The address in the PC is copied to the MAR, the instruction is fetched from that address into the MDR, then copied to the CIR, and the PC is incremented.

○ **C.** The data bus carries the memory address to RAM, the control bus signals a read operation, then the address bus returns the instruction.

○ **D.** The Control Unit generates a random address, the instruction at that address is executed, then the next instruction is selected.

Q17 (2 marks)

A computer has a 32-bit data bus and a 32-bit address bus.
Which statement correctly describes what these specifications mean?

○ **A.** The CPU can transfer 32 bytes of data at once and address up to 32 memory locations

○ **B.** The CPU can transfer 32 bits of data at once and address up to 4 GiB of memory

○ **C.** The CPU can only process data in multiples of 32 and requires 32 RAM chips

○ **D.** The CPU runs at 32 GHz and has 32 registers available

Q18 (2 marks)

A computer has both registers inside the CPU and RAM connected via buses.
Which statement correctly explains why registers are used when RAM already exists?

- ○ **A.** Registers can store larger amounts of data than RAM, so they are used for big files.
- ○ **B.** Registers are non-volatile, so they retain data when the computer is turned off, unlike RAM.
- ○ **C.** Registers are cheaper to manufacture than RAM, so they are used to reduce computer costs.
- ○ **D.** Registers are located inside the CPU and can be accessed much faster than RAM, making them ideal for data the CPU is actively processing.

Q19 (2 marks)

During the decode stage of the fetch-decode-execute cycle, the instruction in the CIR is processed.
Which statement correctly describes what happens during the decode stage?

- ○ **A.** The instruction binary code is converted to human-readable text so the programmer can verify it.
- ○ **B.** The Control Unit interprets the instruction and prepares the necessary control signals for execution.
- ○ **C.** The ALU checks whether the instruction contains any syntax errors before it can be executed.
- ○ **D.** The MDR transfers the instruction to RAM so it can be permanently stored.

Q20 (2 marks)

During the decode stage, the Control Unit examines the instruction in the CIR.
What does the Control Unit do after decoding the instruction?

- ○ **A.** It stores the result of the instruction in the Accumulator
- ○ **B.** It fetches the next instruction from RAM immediately
- ○ **C.** It generates control signals to coordinate the components needed for execution
- ○ **D.** It converts the instruction from binary to hexadecimal for the ALU

Q21 (2 marks)

A computer has a 64-bit data bus connecting the CPU to RAM. Which statement correctly explains the significance of the data bus width?

- ○ **A.** The CPU can address a maximum of 64 different memory locations.
- ○ **B.** Each instruction in the program must be exactly 64 bits in size.
- ○ **C.** The CPU can transfer 64 bits (8 bytes) of data between the CPU and RAM in a single operation.
- ○ **D.** The computer can only run 64-bit operating systems and cannot run 32-bit software.

Q22 (2 marks)

The Memory Data Register (MDR) is described as 'bidirectional'. Which statement correctly explains what this means in terms of the MDR's function?

- ○ **A.** The MDR can hold both positive and negative numbers
- ○ **B.** The MDR can receive data from RAM during reads and send data to RAM during writes

○ **C.** The MDR can communicate with both the MAR and the PC simultaneously

○ **D.** The MDR stores two copies of each piece of data for backup purposes

Q23 (2 marks)

A CPU has a clock speed of 3 GHz.
Which statement correctly explains what this specification means?

○ **A.** The CPU can execute exactly 3 billion complete programs per second

○ **B.** The CPU's clock generates 3 billion pulses per second, synchronising all operations

○ **C.** The CPU has 3 billion transistors inside it

○ **D.** The CPU can store 3 gigabytes of data in its registers

Q24 (3 marks)

A computer system has a 32-bit address bus.
The system designer wants to know the maximum amount of RAM that this system can address.
Which calculation correctly determines the maximum addressable memory?

○ **A.** 32 × 8 = 256 bytes

○ **B.** 2^5 × 2^3 = 256 bytes

○ **C.** 32 × 1024 = 32,768 bytes (32 KiB)

○ **D.** 2^32 = 4,294,967,296 bytes (4 GiB)

Q25 (3 marks)

A computer system designer is choosing between:
• System A: 24-bit address bus
• System B: 28-bit address bus
Which calculation correctly determines how much more memory System B can address compared to System A?

- ○ **A.** System B addresses 4 more memory locations (28 - 24 = 4 extra locations)
- ○ **B.** System B addresses 16 times more memory (2^28 ÷ 2^24 = 2^4 = 16 times more)
- ○ **C.** System B addresses 4 GB more memory (28 - 24 = 4, and each bit adds 1 GB)
- ○ **D.** Both systems address the same amount because they are both measured in bits

Q26 (3 marks)

The following sequence describes operations during the fetch-decode-execute cycle:
1. PC contains address 1000
2. Contents of PC copied to MAR
3. Memory is read from address in MAR
4. Data placed in MDR
5. Contents of MDR copied to CIR
6. PC incremented to 1001
7. CIR decoded by Control Unit
8. Control signals sent to ALU
9. Result stored in Accumulator
Which statement correctly identifies which stages of the cycle are represented by steps 1-6, steps 7, and steps 8-9?

- ○ **A.** Steps 1-6 are Execute, Step 7 is Decode, Steps 8-9 are Fetch
- ○ **B.** Steps 1-6 are Fetch, Step 7 is Decode, Steps 8-9 are Execute

- ○ **C.** Steps 1-6 are Decode, Step 7 is Fetch, Steps 8-9 are Execute
- ○ **D.** Steps 1-6 are Fetch, Step 7 is Execute, Steps 8-9 are Decode

Q27 (3 marks)

A CPU needs to transfer 256 bytes of data from RAM. The system has a 64-bit data bus.
How many separate bus transfers are required to move all 256 bytes?

- ○ **A.** 4 transfers, because 256 ÷ 64 = 4
- ○ **B.** 32 transfers, because 256 bytes ÷ 8 bytes per transfer = 32
- ○ **C.** 256 transfers, because each byte requires its own transfer
- ○ **D.** 2048 transfers, because 256 bytes × 8 bits per byte = 2048 bit transfers

Q28 (3 marks)

The Von Neumann bottleneck is a well-known limitation of the Von Neumann architecture.
Which statement correctly identifies the cause and effect of this bottleneck?

- ○ **A.** The CPU is too slow to process instructions, causing programs to run slowly regardless of memory speed
- ○ **B.** Instructions and data share the same buses, so the CPU cannot fetch the next instruction while accessing data for the current instruction
- ○ **C.** The clock signal cannot reach all components simultaneously, causing timing errors
- ○ **D.** RAM can only store a limited amount of data, forcing programs

to be smaller than the programmer intended

Q29 (3 marks)

The control bus carries various signals between the CPU and other components. These signals coordinate system operations.
A student lists the following as signals carried by the control bus:
1. Memory address
2. Read/Write signal
3. Interrupt request
4. Clock signal
5. Data being transferred
Which signals from the list are correctly identified as control bus signals?

- ○ **A.** 1, 2, and 5 only
- ○ **B.** 2, 4, and 5 only
- ○ **C.** 1, 3, and 4 only
- ○ **D.** 2, 3, and 4 only

Q30 (3 marks)

An embedded system in a washing machine controls the wash cycle, monitors water temperature, and displays status to the user.
Which statement correctly explains why an embedded system is suitable for this application rather than a general-purpose computer?

- ○ **A.** Embedded systems can connect to the internet, allowing remote monitoring of the washing cycle
- ○ **B.** Embedded systems are designed for specific tasks, offering lower cost, lower power consumption, and reliability for dedicated functions
- ○ **C.** Embedded systems use a different type of electricity that is safer near water

○ **D.** Embedded systems can run any software, giving manufacturers flexibility to add new features

Total: **55** marks • Time: **30 min**

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